
4: Keeping Track of Complex Analyses

Workflow management using snakemake

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2025-05-27



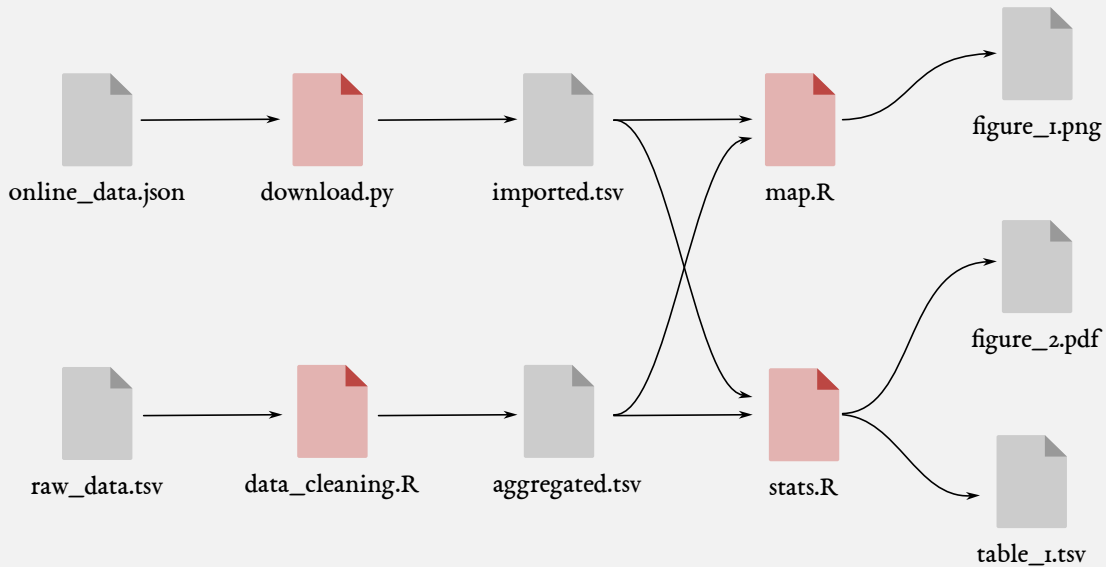
data.tsv



script.R



result.png





readme.md



Image by Simon Kadula



data provenance



re-runability



scalability





data provenance



version control



papertrail deluxe



Image by Jan Kopřiva



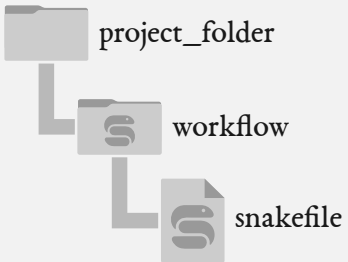
snakefile



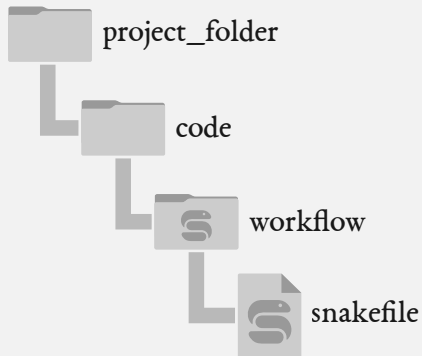
snakefile

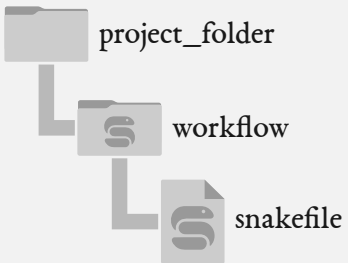


python+

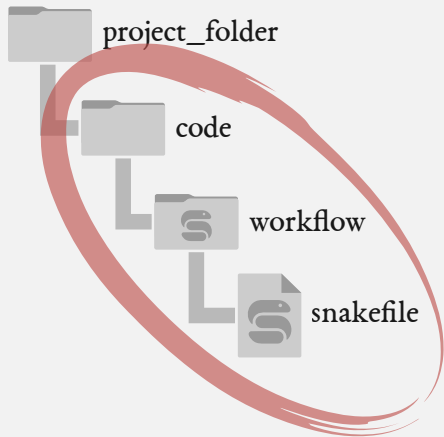


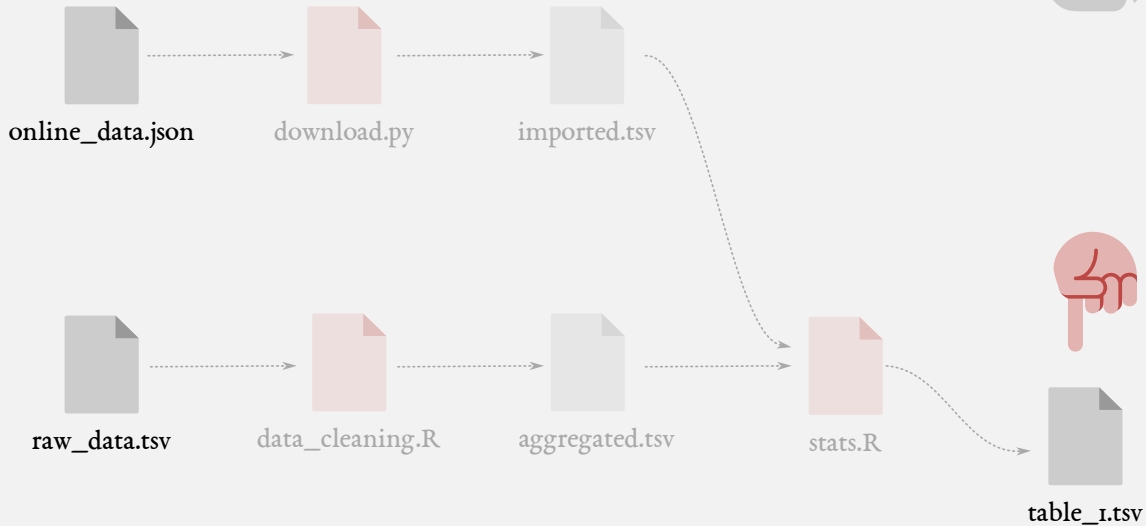
or

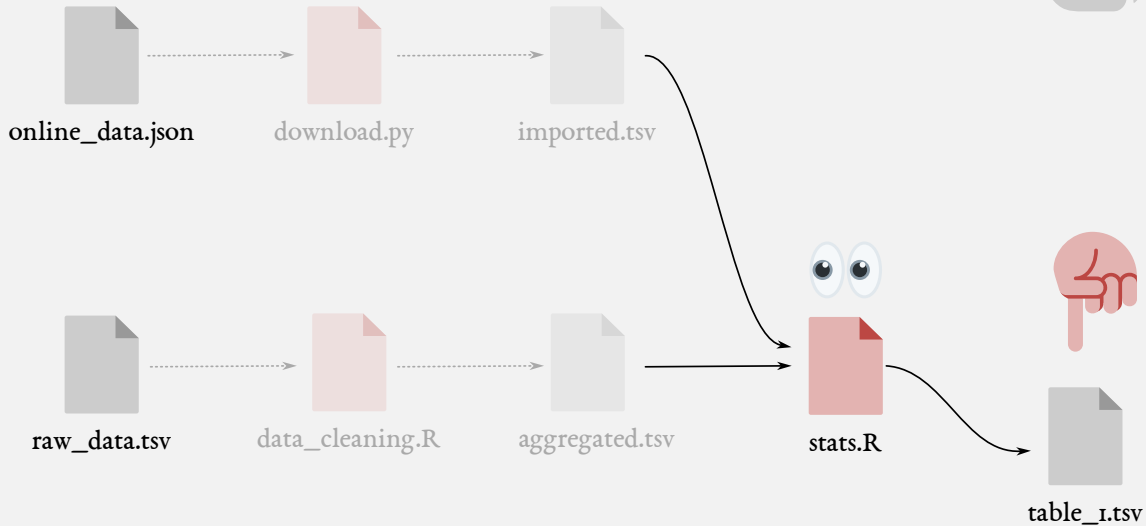


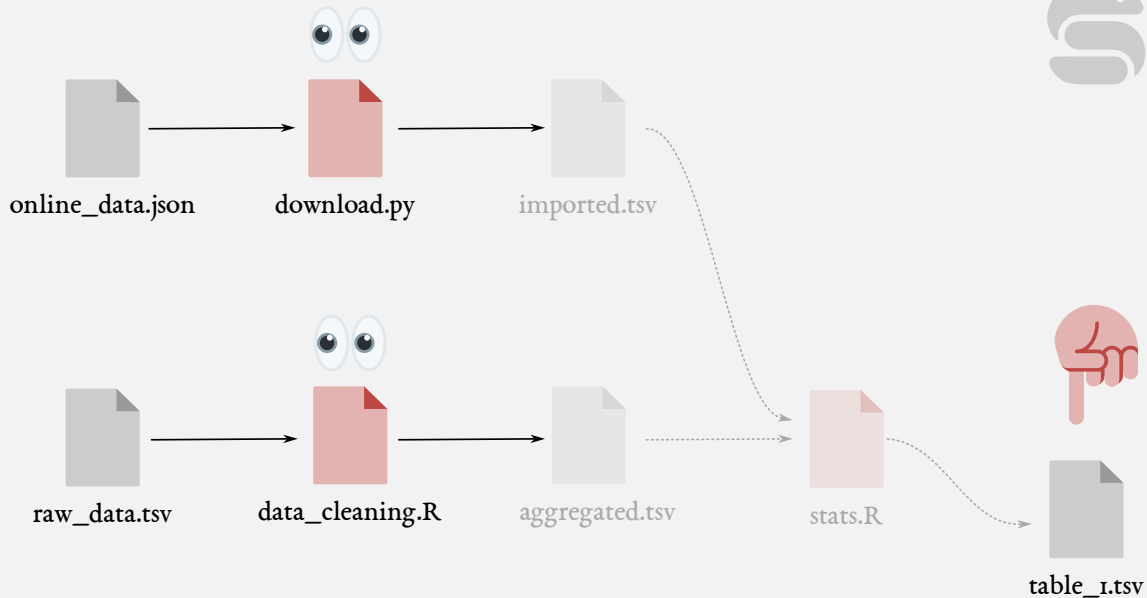


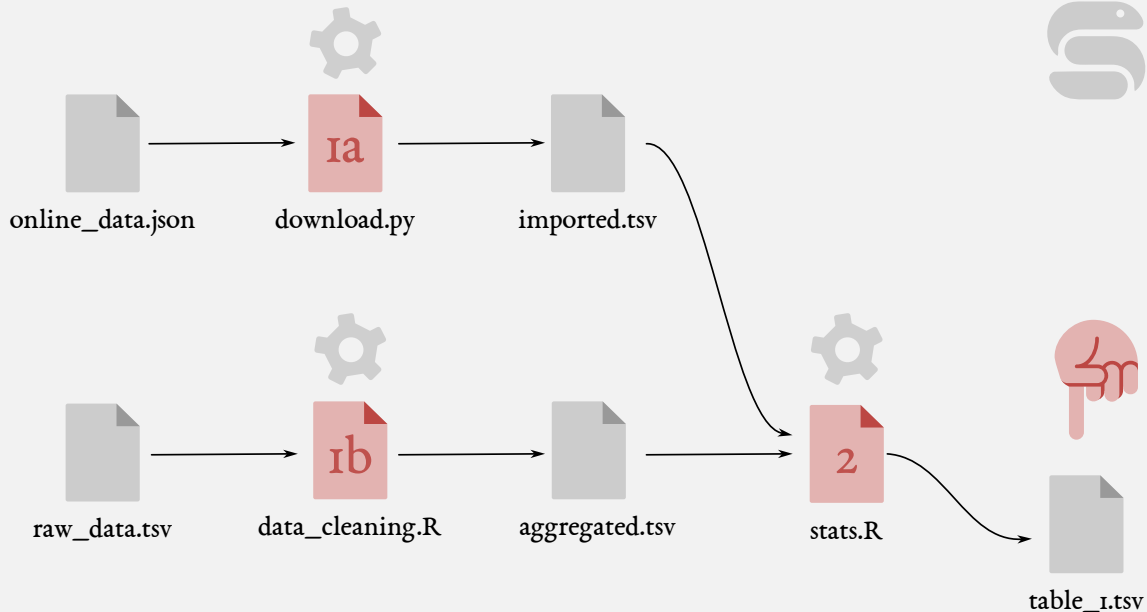
or

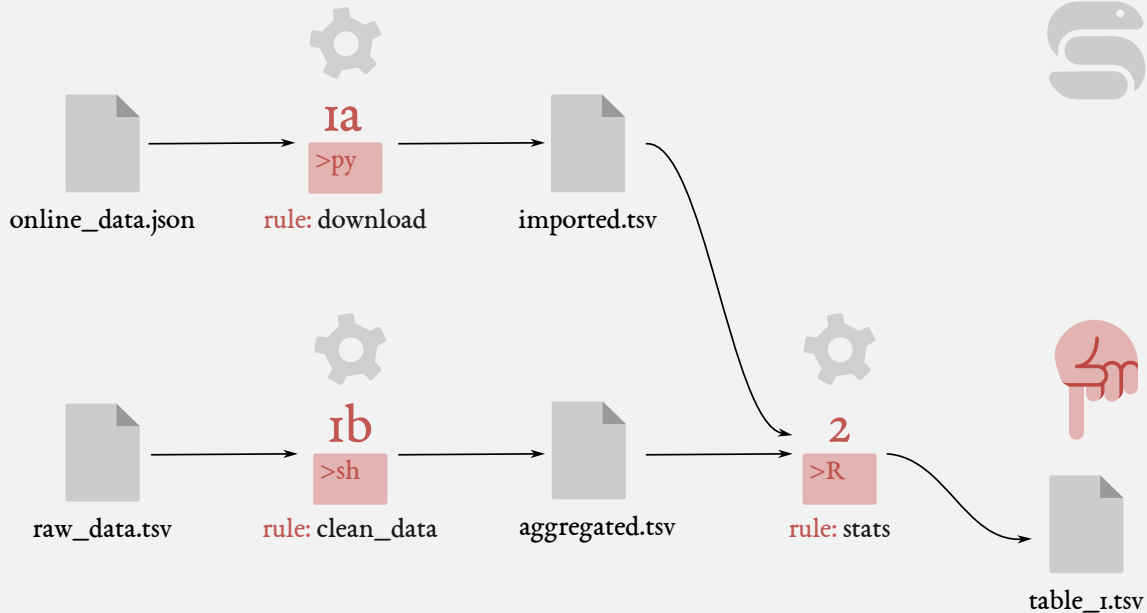














in → **>R** out →

in → **>py** out →

in → **>R** out →



```
rule all:  
  input: "../results/table.tsv"
```

1a

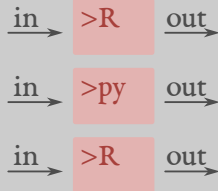
```
rule clean_data:  
  input: "../data/raw_data.tsv"  
  output: "../results/aggregated.tsv"  
  shell:  
    ""  
    Rscript --vanilla R/data_cleaning.R  
    ""
```

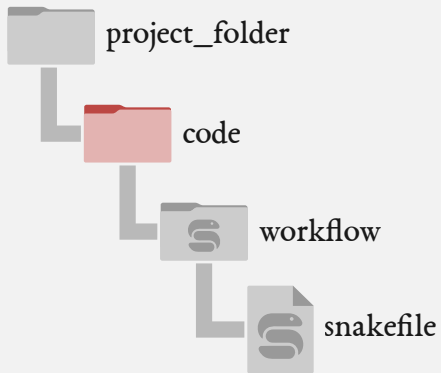
1b

```
rule download:  
  output: "../results/imported.tsv"  
  shell:  
    ""  
    python py/download.py  
    ""
```

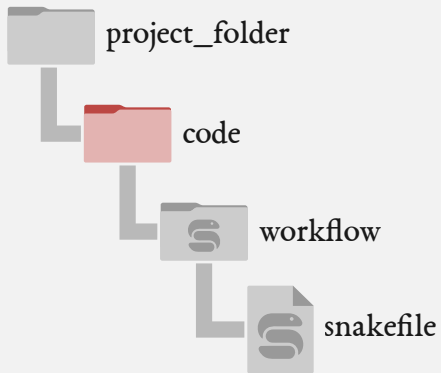
2

```
rule stats:  
  input:  
    "../results/aggregated.tsv",  
    "../results/imported.tsv"  
  output: "../results/table.tsv"
```

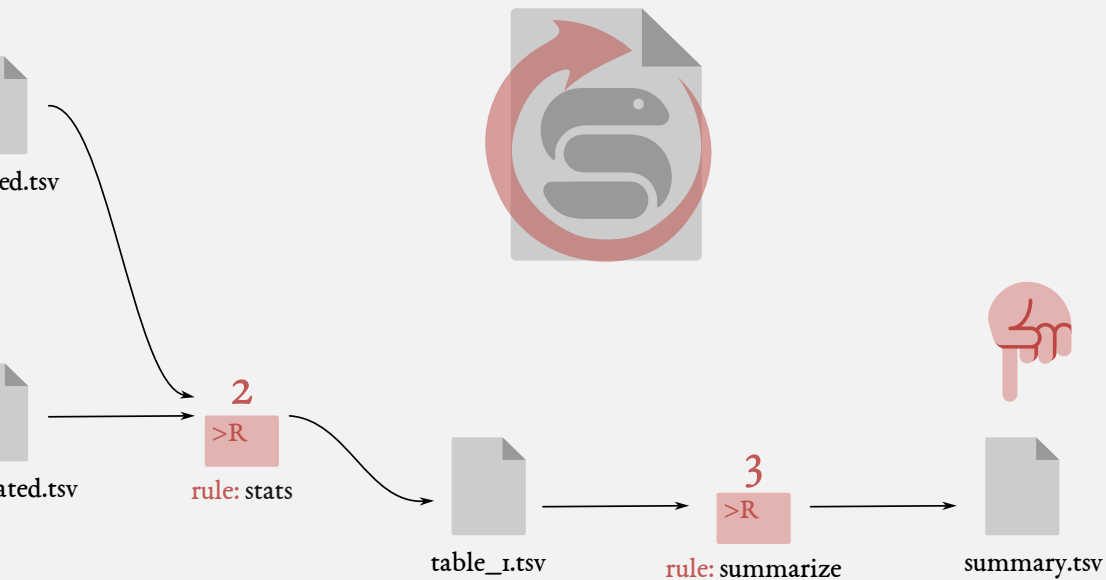


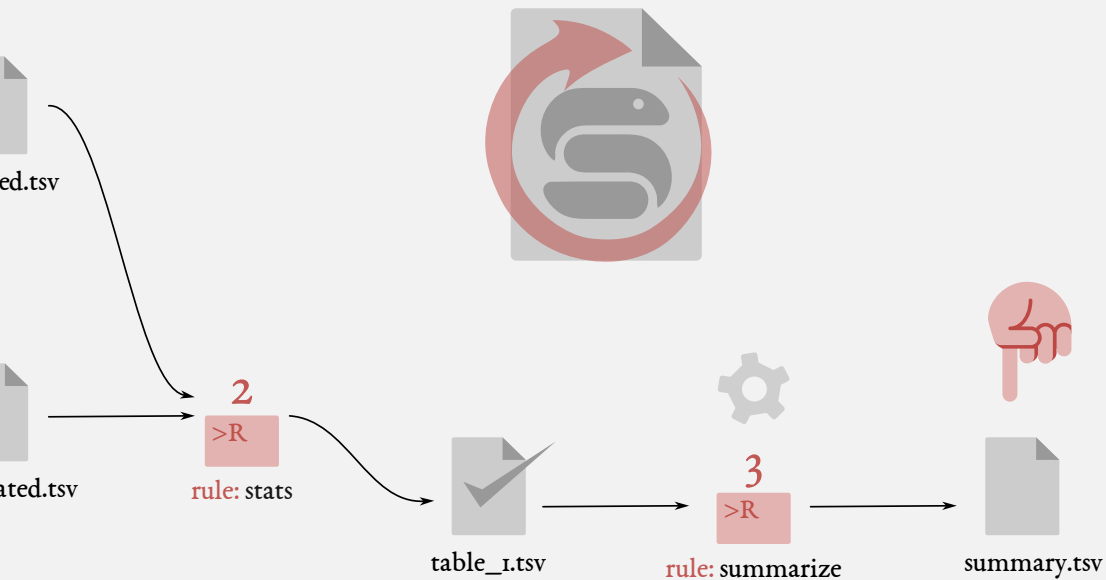


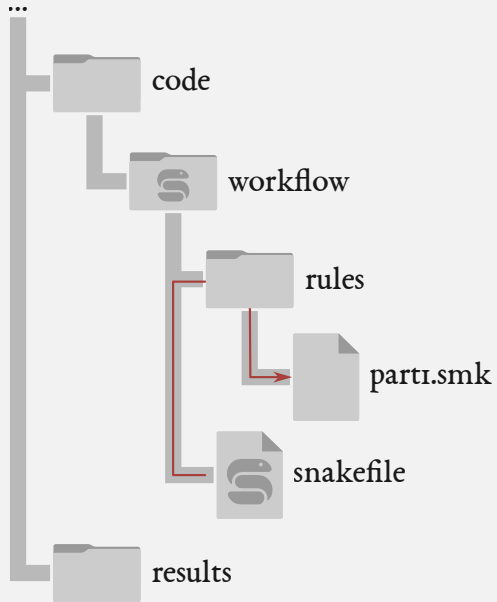
```
code$ snakemake -c 1
```



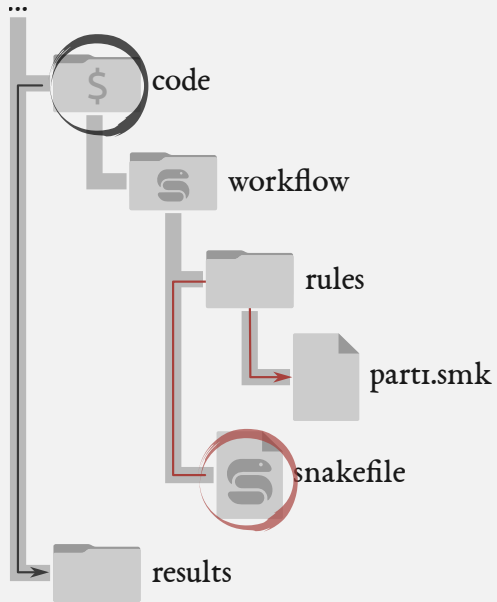
```
code$ snakemake --jobs 5 --cluster '...'
```







```
rule all:  
  input: "../results/final.txt"  
  
  include: "rules/part1.smk"
```



DFG



DFG-SCHWERPUNKTPROGRAMM 1158

ANTARKTIS FORSCHUNG

mit vergleichenden Untersuchungen in arktischen Eisgebieten



created using dalle3